

Course ID and Title: CSCI 270: Introduction to Algorithms and Theory of Computing

Units: 4

Term—Day—Time: [Fall 2026] — [MW 3:50 – 4:50 pm]

Location: THH 201

Instructor: Jiapeng Zhang

Office Hours: Monday 5:00 - 7:00 pm (GCS Floor LL2, ROOM: SB10).

Contact Info: jiapengz@usc.edu

TA: Matin Amini

Office Hours: Friday 2:00 - 4:00 pm (GCS Floor LL2, ROOM: SB7).

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TA: Soumita Hait

Office Hours: Monday 1:00 - 3:00 pm (Location: TBA).

Contact Info: hait@usc.edu

Catalogue Description

Algorithms, including greedy algorithms, divided and conquer, dynamic programming, network flow and some complexity theory.

Course Description

Algorithms are the heart of computer science. It provides mathematically rigorous solutions to computational problems. This course introduces basic methods for the design and analysis of algorithms. It includes greedy algorithms, divide and conquer, dynamic programming and maximum-flow problems. In the last part of this course, we will also discuss topics in the theory of computing. Materials such as decidability/undecidability, polynomial-time reductions, P v.s. NP, and NP-completeness will be covered.

Learning Objectives

The course is designed to enable students to:

1. Understand basis algorithms such as greedy algorithms, divide and conquer, dynamic programming and graph problems.
2. Learn how to model computational problems as mathematical problems, and how to analyze algorithms rigorously.
3. Understand basic concepts from the computation theory such as P and NP.

Recommended Preparation:

Knowledge of the material covered in CSCI 104 and CSCI 170 (or the equivalent). If you do not formally have this knowledge and still wish to take this course, please send me an email to discuss whether you should enroll in this course.

Course Notes:

Grading includes two parts:

- Homework (0%): There will be homework assignments roughly every week (a total of about 10-11). The problems are intended to be challenging, so plan ahead.
- Midterm Exam (60%): The midterm will cover the class material up to that point.
- Final Exam (40%): The final will cover the material of the entire semester, though more focus will be on the second half of the semester.

Grading Breakdown:

Assessment Tool (assignments)	Points	% of Grade
Homework	0	0%
Midterm exam	60	60%
Final exam	40	40%
TOTAL	100	100%

Textbook:

- *Algorithm Design*, by Jon Kleinberg and Éva Tardos.

Participation

Participation is not required

Grading Scale

Percentage	Grade
90%	A
85%	A-
80%	B+
75%	B
70%	B-
65%	C+
60%	C
55%	C-
50%	D
<50%	F

Assignment Submission Policy:

Homework assignments will be submitted by Gradescope.

Grading Timeline:

Grading of each homework will be ready one week after submission.

Use of Generative AI in this Course: There is no restriction of using generative AI.

Generative AI is encouraged: You are expected to use AI (e.g., ChatGPT and image generation tools) in this class. Learning to use AI is an emerging skill; this is an opportunity for you to discuss appropriate use of these tools with the instructor. Keep in mind the following:

- AI tools are permitted to help you brainstorm topics or revise work you have already written.
- If you provide minimum-effort prompts, you will get low-quality results. You will need to refine your prompts to get good outcomes. This will take work.
- Proceed with caution when using AI tools and do not assume the information provided is accurate or trustworthy. If it gives you a number or fact: assume it is incorrect unless you either know the correct answer or can verify its accuracy with another source. You will be responsible for any errors or omissions provided by the tool. It works best for topics you understand.
- AI is a tool, but one that you need to acknowledge using. Please include a paragraph at the end of any assignment explaining if, how, and why you used AI and indicate/specify the prompts you used to obtain the results. Failure to do so is a violation of academic integrity policies.

Course Schedule

	Topics/Daily Activities	Readings/Preparation	Deliverables
Week 1	Introduction, Stable Matching	Textbook: Section 1,2,3	
Week 2	Divide and Conquer	Textbook: Section 5	
Week 3	Divide and Conquer	Textbook: Section 5	
Week 4	Divide and Conquer	Textbook: Section 5	
Week 5	Dynamic Programming	Textbook: Section 6	Midterm 1
Week 6	Dynamic Programming	Textbook: Section 6	
Week 7	Dynamic Programming	Textbook: Section 6	
Week 8	Greedy Algorithms	Textbook: Section 4	
Week 9	Greedy Algorithms	Textbook: Section 4	Midterm 2
Week 10	Greedy Algorithms	Textbook: Section 4	
Week 11	Max Flow / Min cut	Textbook: Section 7	
Week 12	Max Flow / Min cut	Textbook: Section 7	
Week 13	NP-Completeness	Textbook: Section 8	Midterm 3
Week 14	NP-Completeness	Textbook: Section 8	
Week 15	Computability theory	Textbook: Section 9	
FINAL	Final Exam	Final Exam	Refer to the final exam schedule in the USC <i>Schedule of Classes</i> at classes.usc.edu .

Academic Integrity

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct is contrary to this fundamental mission and includes any act of dishonesty in the production or submission of academic work (either in draft or final form), as well as cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage. Students are expected to uphold the highest standards of academic integrity in all coursework.

This course follows the expectations for academic integrity as stated in the [USC Student Handbook](#). All students are expected to submit assignments that are their own original work and prepared specifically for this course and section in this academic term. Students may not submit work written by others or “recycle” work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of academic misconduct will be reported to the Office of Academic Integrity.

Academic dishonesty has a far-reaching impact and is considered a serious offense against the university. Violations will result in a grade penalty, such as a failing grade on the assignment or in the course, and disciplinary action from the university, such as suspension or expulsion.

For more information about academic integrity see the [Student Handbook](#), the [Office of Academic Integrity's website](#), and university policies on [Research and Scholarship Misconduct](#).

Students should ask their instructor when unsure what constitutes unauthorized assistance on an exam or assignment or what information requires citation and/or attribution.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Student Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on the academic freedom of other students as well as the instructor. ([Living our Unifying Values: The USC Student Handbook](#), page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposes other than individual or group study. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which had been distributed to students or in any way had been displayed for use in relation to the class, whether obtained in class, via email, on the internet, or via any other media. Distributing course material without the instructor's permission will be presumed to be an intentional act to facilitate or enable academic dishonesty and is strictly prohibited. ([Living our Unifying Values: The USC Student Handbook](#), page 13).

Statement on University Academic and Support Systems

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. [The Office of Student Accessibility Services](#) (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Student Financial Aid and Satisfactory Academic Progress:

To be eligible for certain kinds of financial aid, students are required to maintain Satisfactory Academic Progress (SAP) toward their degree objectives. Visit the [Financial Aid Office webpage](#) for [undergraduate](#)- and [graduate-level](#) SAP eligibility requirements and the appeals process.

Support Systems:

[Counseling and Mental Health](#) - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

[988 Suicide and Crisis Lifeline](#) - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline consists of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

[CARE-SC: Confidential Advocacy, Resources, and Education Support Center](#) - (213) 740-9355(WELL) – 24/7/365 on call

Confidential advocates, prevention educators, and professional counseling teams work to promote a universal culture of consent, as well as prevent and respond to sexual assault, intimate partner violence, stalking, or other relationship harm. Services available to all USC students at no cost.

[Office of Civil Rights Compliance](#) - (213) 740-5086

Information about how to get help or help someone affected by harassment, discrimination, retaliation on the basis of a protected characteristic, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

[USC Report & Response](#) - (213) 740-2500

The USC Report & Response website is the university's central reporting portal for concerns arising in the academic space or workplace. All concerns will be assessed and referred to the appropriate university office for resolution. Any questions about USC Report & Response or reporting, in general, can be referred to the [Office of Professionalism and Ethics](#) at ope@usc.edu.

[USC Campus Support and Intervention](#) - (213) 740-0411

Focuses on student success by assisting students in navigating and resolving complex issues through problem solving, presenting options, and connecting to resources.

[USC Emergency Information](#)

Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

[USC Department of Public Safety](#)

For 24 hour emergency assistance or to report a crime: UPC: (213) 740-4321, HSC: (323)-442-1000.

For 24 hour non-emergency assistance or information: UPC: (213) 740-6000, HSC: 323-442-1200.

[Office of the Ombuds](#) - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

[Occupational Therapy Faculty Practice](#) - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.